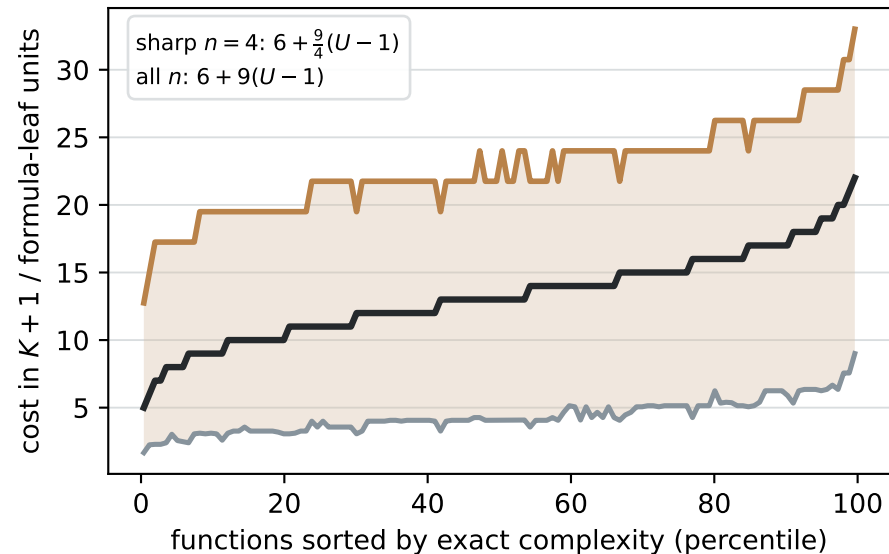


Semantic complexity envelopes: general bounds and sharp finite constants

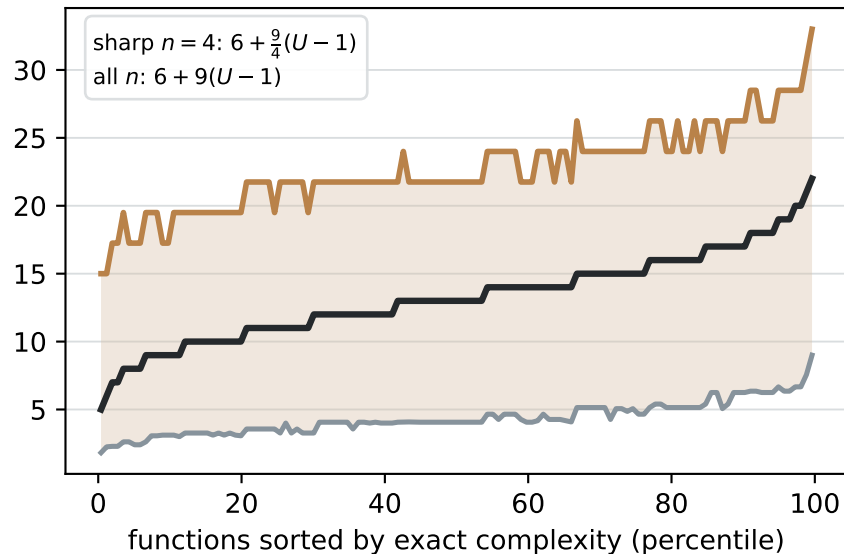
Solid upper curves are exhaustive four-input envelopes; the second formula in each panel holds for every input dimension.

— general lower $B(f)$ — exact $K_L(f) + 1$ — sharp $n = 4$ upper - - - tree component prime-cover $Q_{\min}$ shaded verified four-bit interval

NAND



NOR



AND/OR/NOT

